

KPP Validation Report

MITgcm vs Python Port Comparison

Generated: 2026-09-27 05:58:25

MITgcm Reference:

File: mitgcm_kpp_outputs_lab_sea_1000_0820T0946.nc
UUID: 67a7e785-3230-4ff9-ae78-aba58fe9d581

Python Port Output:

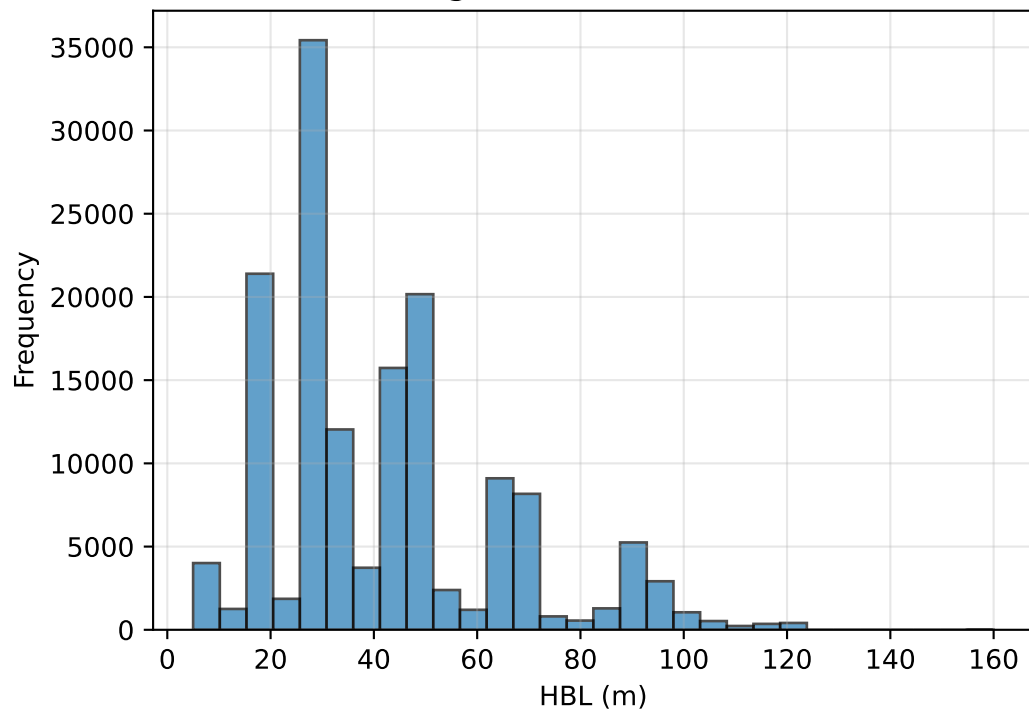
File: python_kpp_outputs_lab_sea_1000_0820T0946.nc
UUID: 67a7e785-3230-4ff9-ae78-aba58fe9d581

Provenance: ☒ UUIDs match

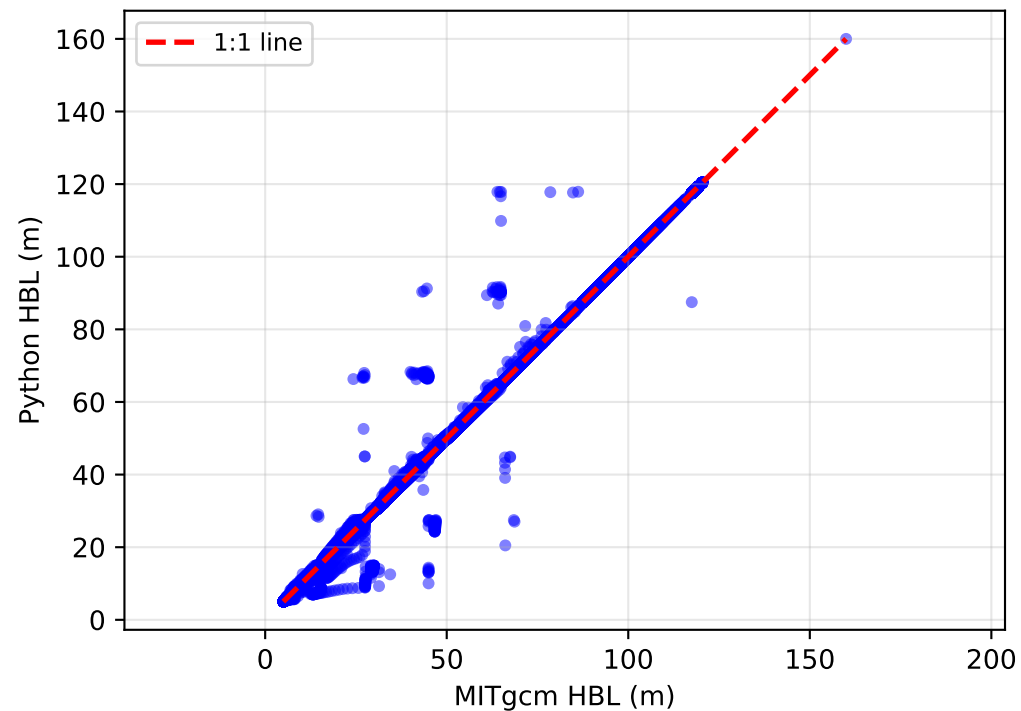
Timesteps: 999
Grid: 20 × 16 × 23

Boundary Layer Depth (HBL) Comparison

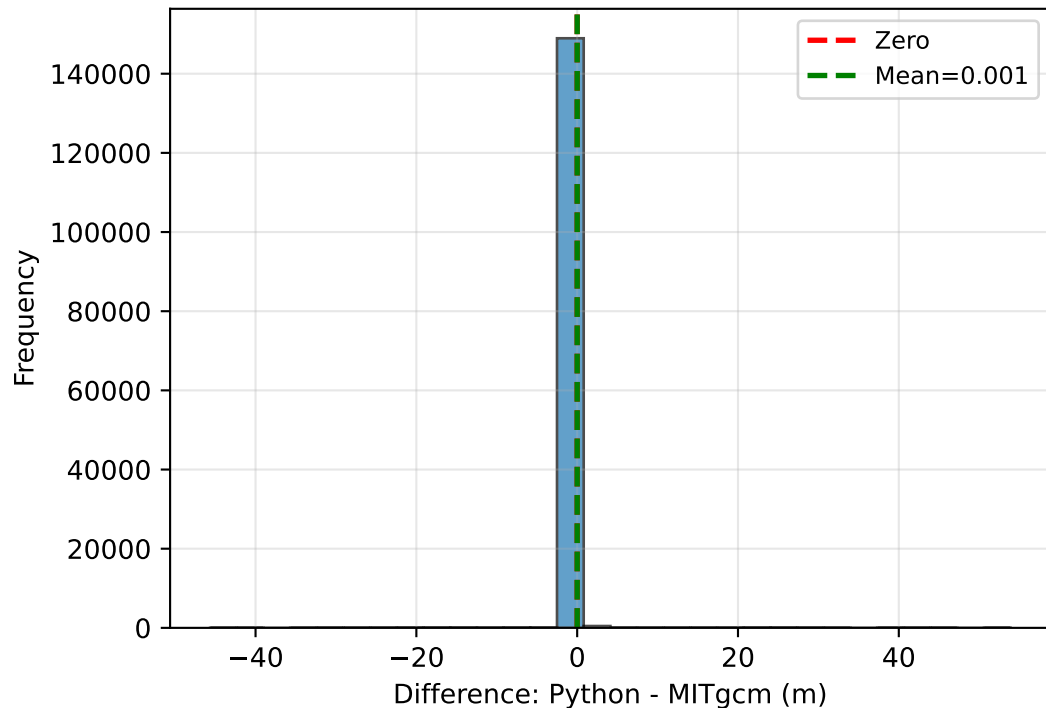
MITgcm HBL Distribution



HBL Scatter Plot



HBL Difference Distribution



HBL Statistics:

Mean |diff|: 0.080266 m
Max |diff|: 53.908091 m
RMS diff: 1.036141 m

Mean rel error: 0.2838%
Median rel error: 0.0083%

Points compared: 149850

Range:

MITgcm: 5.0 – 160.0 m
Python: 5.0 – 160.0 m

Assessment:

△ CHECK

Mixing Coefficients Statistics

Active Mixing Regions (coefficient > 1e-6 m²/s)

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VISC_AZ:	□ EXCELLENT
Median relative error:	0.000000%
Mean relative error:	33.138446%
Max relative error:	173184.718323%
95th percentile error:	22.930534%
Median absolute diff:	0.000000e+00 m ² /s
Points compared:	1637981
DIFF_KZ_S:	□ EXCELLENT
Median relative error:	0.000000%
Mean relative error:	45.234991%
Max relative error:	491254.129038%
95th percentile error:	15.007880%
Median absolute diff:	0.000000e+00 m ² /s
Points compared:	1637981
DIFF_KZ_T:	□ EXCELLENT
Median relative error:	0.000000%
Mean relative error:	45.234991%
Max relative error:	491254.129038%
95th percentile error:	15.007880%
Median absolute diff:	0.000000e+00 m ² /s
Points compared:	1637981
GHAT:	□ GOOD
Median relative error:	0.132176%
Mean relative error:	0.734684%
Max relative error:	774.559456%
Points compared:	391574

Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: ☒ NEEDS REVIEW

COMPONENT RESULTS:

HBL (Boundary Layer Depth):	☒ CHECK
Max difference:	53.908091 m
Mean relative error:	0.2838%
VISC_AZ:	☐ EXCELLENT
Median relative error:	0.000000%
DIFF_KZ_S:	☐ EXCELLENT
Median relative error:	0.000000%
DIFF_KZ_T:	☐ EXCELLENT
Median relative error:	0.000000%
GHAT:	☐ GOOD
Median relative error:	0.132176%

PROVENANCE:

UUID Match:	☐ Yes
MITgcm UUID:	67a7e785-3230-4ff9-ae78-aba58fe9...
Python UUID:	67a7e785-3230-4ff9-ae78-aba58fe9...

ACCEPTANCE CRITERIA:

☐ EXCELLENT:	HBL < 0.1m, mixing < 0.1% error
☐ GOOD:	HBL < 1.0m, mixing < 1.0% error
☒ CHECK:	Exceeds thresholds - investigate